Growth Function

Suppose $h \in H$ maps X to $\{-1, 1\}$

$x_1, x_2, \dots x_N \in X$ gives

$h(x_1) \dots h(x_N)$ a N tuple of ± 1

called DICHOTOMY, it splits

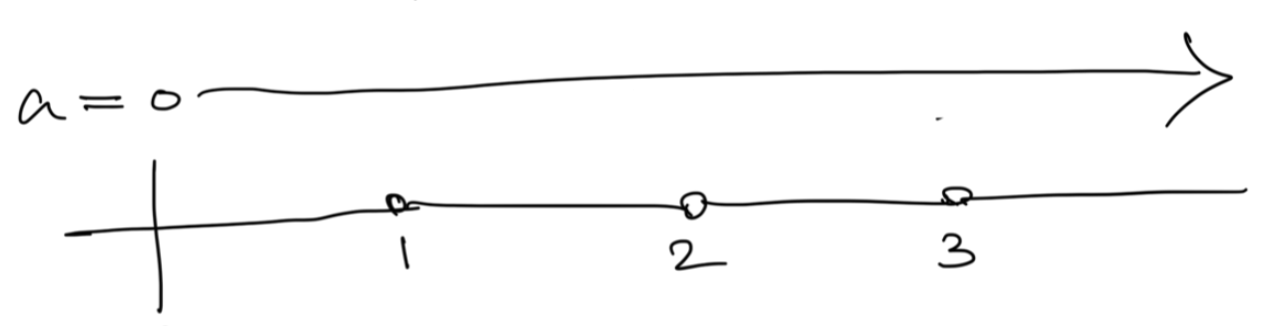
$x_1 \dots x_n$ to two groups.

Each h generates a dichotomy but different h can give same pattern of ± 1 on a sample.

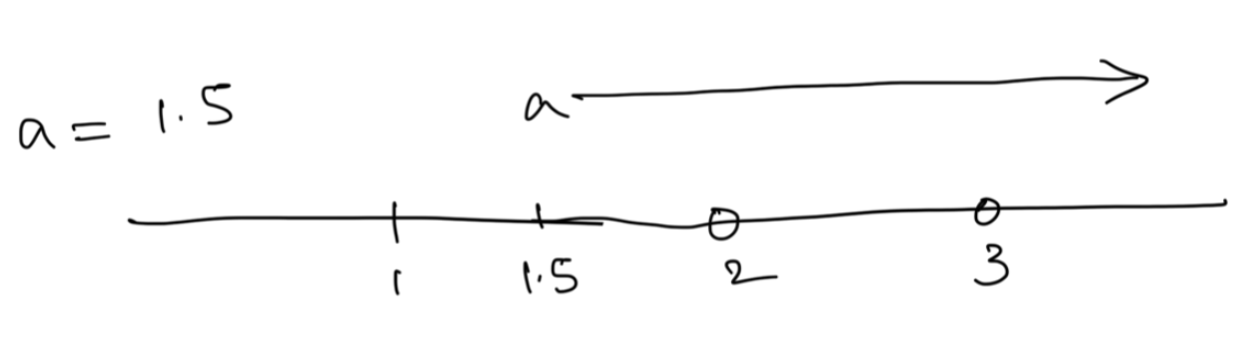
Example: $N=3$

$x_1 = 1 \quad x_2 = 2 \quad x_3 = 3$

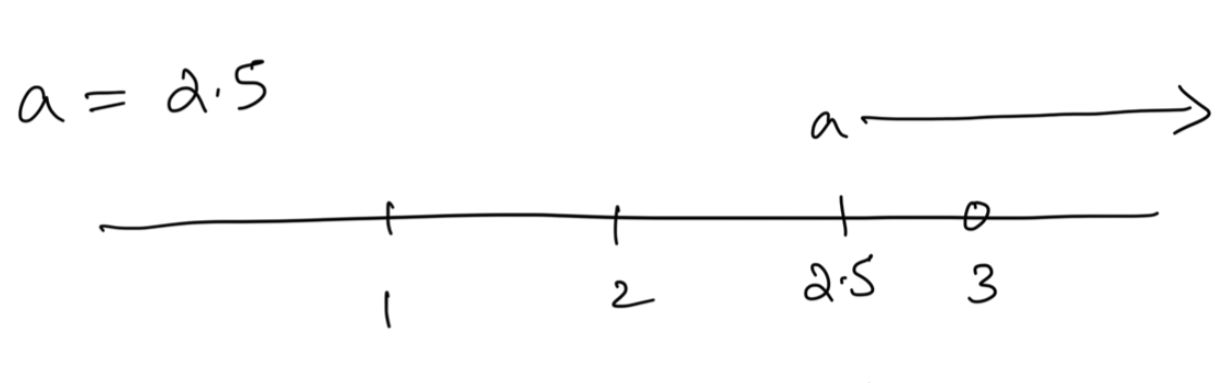
total possibilities $2^N = 2^3 = 8$



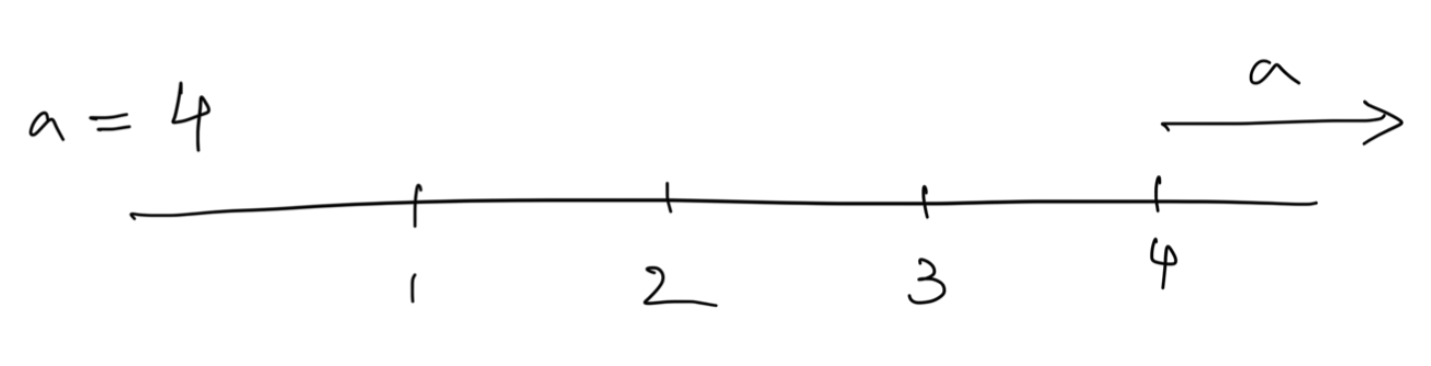
$(+1, +1 \quad +1)$



$(-1, +1, +1)$



$(-1, -1, +1)$



$(-1, -1, -1)$

There are infinitely many hypothesis but only 4 different dichotomies on these three points

$H(1,2,3) = \{ (+++), (-++), (--+), (---) \}$

The growth function $m_H(N)$ is

$$\max_{x_1, x_2 \dots x_m \in X} |H(x_1, x_2 \dots x_N)|$$

where $|\cdot|$ denotes cardinality of set

$\Rightarrow m_H(N)$ = maximum no of dichotomies that can be generated by H on N points. We consider all possible choices of N points $x_1, \dots x_N$ and pick one that gives most dichotomies

For any H, $H(x_1, \dots x_N) \subseteq \{-1, 1\}^N$

$$m_H(N) \leq 2^N$$

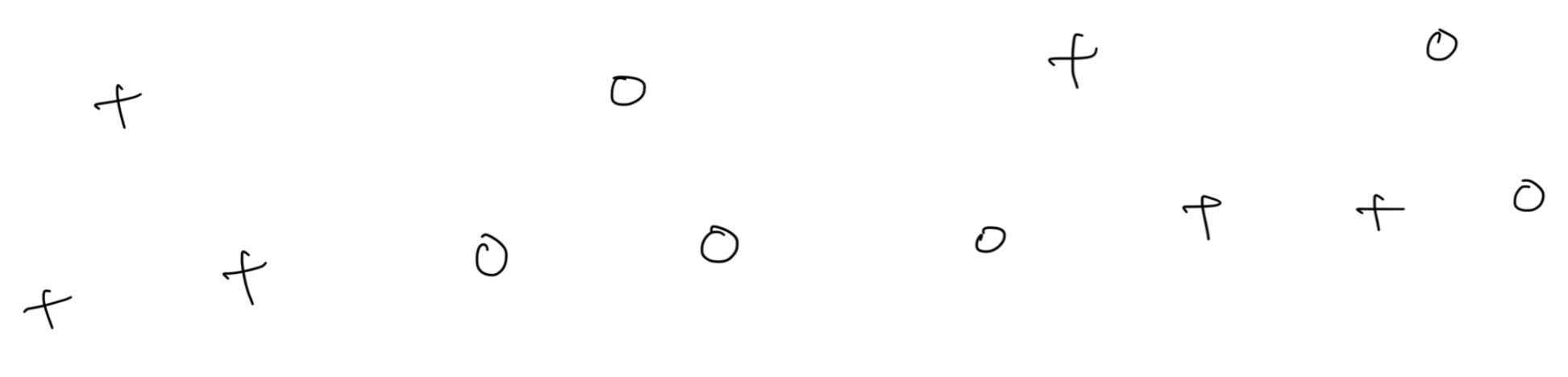
if H is capable of generating all possible dichotomies on $x_1 \dots x_N$ then H can SHATTER $x_1 \dots x_N$

a) For N points in a line, a threshold can be placed in $N+1$ regions, so

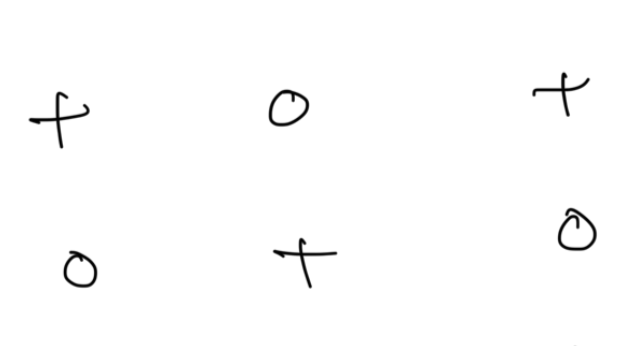
$m_H(N) = N+1$, different dichotomies

b) For a 2D perceptron with $N=3$

1) if points are non collinear we have 2^3 possible dichotomies and we can generate all



2) if they are collinear



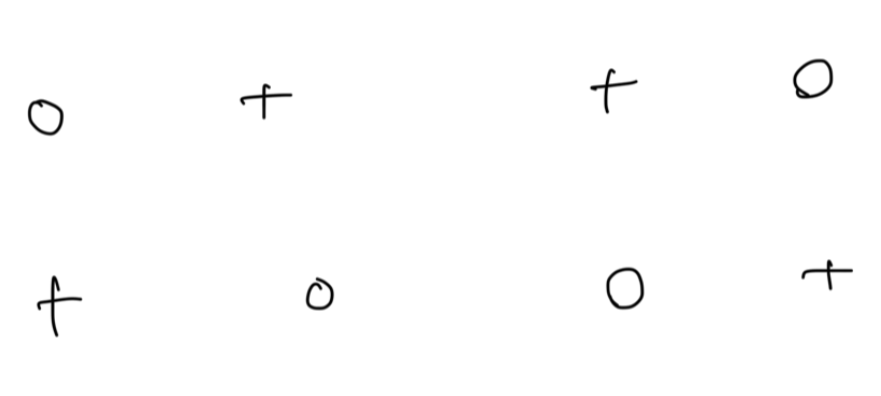
this arrangement cant be seperated by a perceptron, so only 6 possible

$$m_H(N) = \max_{x_1 \dots x_N} |H(x_1 \dots x_N)|$$

$$= 8$$

we are allowed to choose best arrangement of points

For $N=4$



$2^4 - 2 = 14$ possible dichotomies

let $x_1, x_2, x_3, x_4 \in R^2$

$x_1 = \begin{bmatrix} x_{11} \\ x_{12} \end{bmatrix} \quad x_2 = \begin{bmatrix} x_{21} \\ x_{22} \end{bmatrix} \quad \dots$

$x_1 = (x_1, x_2) \longrightarrow \tilde{x}_1 = (x_1, x_2, 1)$

$\tilde{x}_i = \begin{bmatrix} x_{i1} \\ x_{i2} \\ 1 \end{bmatrix}$

$w = \begin{bmatrix} w_1 \\ w_2 \\ -b \end{bmatrix}$

$w^T \tilde{x}_1 = [w_1 \; w_2 \; b] \begin{bmatrix} x_{i1} \\ x_{i2} \\ 1 \end{bmatrix}$

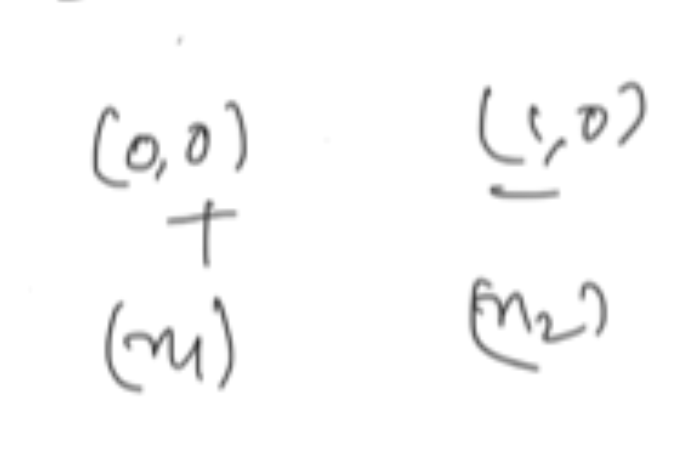
$= w_1 x_{i1} + w_2 x_{i2} + b$

$h(x_i) = \text{sign}(w^T \tilde{x}_i)$

We have 4 vectors in 3 dimensions, the vectors must be linearly dependent

(ie) $\alpha_1, \alpha_2, \alpha_3\, \alpha_4$ exists such that

$\alpha_1 \tilde{x}_1 + \alpha_2 \tilde{x}_2 + \alpha_3 \tilde{x}_3 + \alpha_4 \tilde{x}_4 = 0$



$\tilde{x}_1 = (0,0,1)$

$\tilde{x}_2 = (1,0,1)$

$\tilde{x}_3 = (0,1,1)$

$\tilde{x}_4 = (1,1,1)$

$\tilde{x}_1 - \tilde{x}_2 - \tilde{x}_3 + \tilde{x}_4 = 0$

$\alpha_1 = 1, \alpha_2 = -1, \alpha_3 = -1, \alpha_4 = 1$

$(+, -, -, +)$

$w^T \tilde{x}_1 > 0$ when $x_1 > 0$

$w^T \tilde{x}_2 < 0$ when $x_2 < 0$

$w^T x_3 < 0$

$w^T x_4 > 0$

$\alpha_1 w^T \tilde{x}_1 + \alpha_2 w^T \tilde{x}_2 + \alpha_3 w^T \tilde{x}_3 + \alpha_4 w^T \tilde{x}_4 > 0$

$(+)(+) \quad (-)(-) \quad (-)(-) \quad (+)(+)$

But $\sum \alpha_i w^T \tilde{x}_i = 0$

$\therefore (+, -, -, +)$ cannot be solved

Similarly $(-, +, +, -)$

For linear classifiers in d dimension the growth function (Covers Theorem)

$$m_H(N) = 2 \sum_{k=0}^{d} {}^{N-1}C_k$$

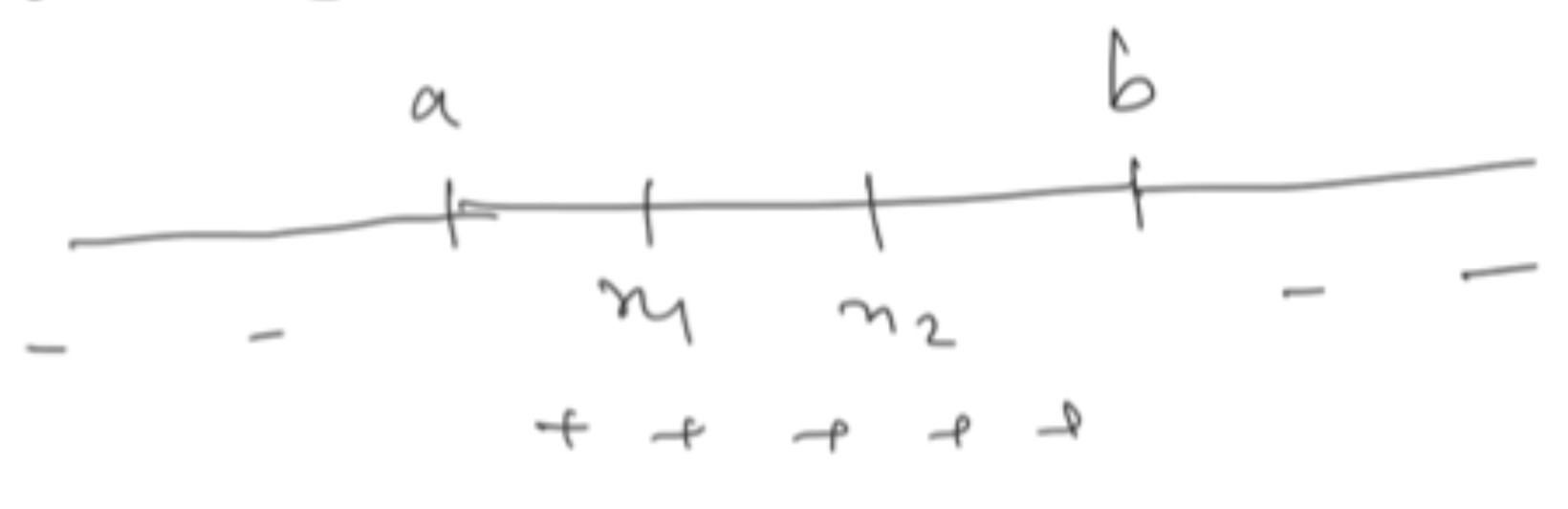
For $d=2, N=4$

$$m_H(4) = 2 \left[{}^3C_0 + {}^3C_1 + {}^3C_2 \right]$$

$$= 2[1 + 3 + 3]$$

$$= 14$$

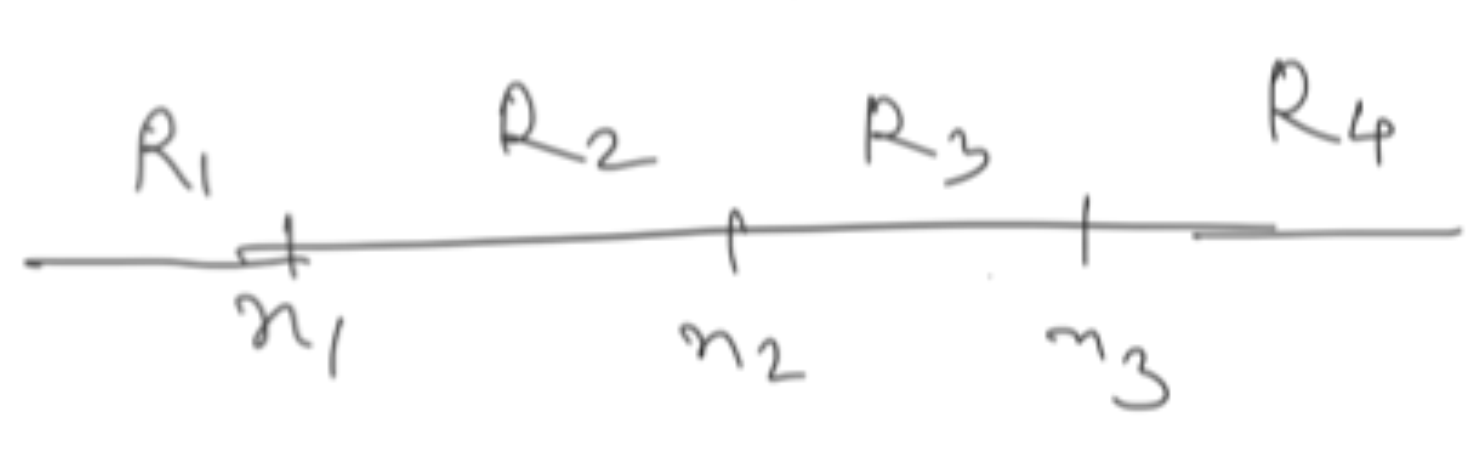
c) Positive Intervals



We need

$$0 \leq R_a \leq R_b \leq N$$

For $N=3$



$(+ - -) \quad (- - -)$

$(- + -) \quad (+ + -)$

$(- - +) \quad (- + +)$

$(+ + +)$

$(+ - +)$ is not possible

${}^{N+1}C_2$ = choose 2 different regions from $N+1$

$N+1$ = where a, b are same endpoints

$(a,b) \in R_1$ or $(a,b) \in R_2$

$(a,b) \in R_3$ or $(a,b) \in R_4$

But these 4 produce $(-,-,-,-)$ labeling, only one dichotomy

So total no of dichotomy is

$${}^{N+1}C_2 + 1 = m_H(N)$$

d) Convex Sets

$$h(x) = \begin{cases} +1, & x \in C \\ -1, & x \notin C \end{cases}$$

if we choose all N points on a circumference of a circle, connect $+1$ points with a polygon we get a convex hull, as all N points lie on circumference of circle, no (-1) point can lie inside convex hull of $(+1)$ points

Every possible dichotomy of (N) points can be realised by some convex sets

$$m_H(N) = 2^N$$